



Carbon fiber blades of a drone

Google something pointless like “funny cat videos”. Incidentally, your smartphone’s USB-C charger likely has more computing power than the Apollo spacecraft, because it has a chip that smart-charges your phone...

On a quantum scale such as when talking about all of nanotechnology and its implementations, strange effects start to show up. This is because the quantum effects of atoms and atomic particles (electrons for example) start showing up at the tiniest scale. For example, we all know that copper has a particular colour, and is opaque, right? Not at the nano-scale, where a thin layer of copper is actually transparent!. Most known to Digit readers is the effect of an insulator like silicon becoming a conductor of electricity at a nano-scale, which is the very basis for all of the technology we have, and why silicon is called a semiconductor.



### NANO NOW (THE CURRENT STATE OF NANOTECHNOLOGY)

While its use in tech is what springs to mind for all of us in the Digit family, nanotechnology is far more pervasive and ubiquitous than merely inside CPUs, RAM or silicon. And, of course, this article is only interested in human-engineered nano-particles and nanotechnology,

because otherwise, in nature, every atom around us qualifies as a nano-particle... and the study of that is quite simply, physics.

One of the most common uses of nanotechnology is the creation of artificial materials that are often tougher and lighter than anything natural. Carbon fibre is the most common example of this, and it is used in pretty much every manufacturing vertical in some way or the other. Everything from shoes to tennis racquets, cars to airplanes, missiles to speakers, and medical implants to mechanical parts, all have carbon fibre-based materials used in them. We’ll come back to this later when we talk about research in the field of nanomaterials.

What we’re getting at is that nano-engineering is all around us, and is being used to develop better drugs to fight ailments such as cancer, diabetes, COVID-19 and pretty much anything else, to creating better batteries to store more energy. Every single one of us who uses spectacles is probably already using nanotechnology – if you got a coating added on to your spectacles for anti-glare. The phone screens that cause many of us to need spectacles are also using nanotech to create scratch resistant and shatterproof glass, as do the screen-guards we overlay on top of those precious screens of ours.

Unless you organically grow your own food, you’re probably already eating nanotech! A lot of foods such as packaged yoghurt use nanoparticles of titanium dioxide to make the yoghurt look pristine and white – those who make yoghurt at home will know what we’re talking about. Not only does food colouring like titanium dioxide make the food look

#### Medical bioimaging

Innovative approaches of folding DNA into functional structures has opened up new possibilities of using DNA based nanotechnologies for medical imaging and sensing purposes.  
<http://dgit.in/nanoprobe>

#### Industry in India

The steps taken by the Government of India towards encouraging nanotechnology research and industries, as well as a review of the current state of the field in the country.  
<http://dgit.in/nanohere>

#### Implants

In the current state, nanotechnologies are most used for medicinal purposes, here is how specific biocompatible materials are being developed for medical implants.  
<http://dgit.in/nanoimp>